

Vueports Solutions — Deployment & Configuration Guide

Version: 1.0 | Last Updated: 2026-05-27 | Target: Ubuntu 22.04 LTS + MySQL 8.0 + PHP 8.2

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1. Pre-Deployment Checklist

Minimum Requirements

Resource	Minimum	Recommended
OS	Ubuntu 20.04 LTS	Ubuntu 22.04 LTS
CPU	2 vCPU	4 vCPU
RAM	2 GB	4 GB
Storage	40 GB SSD	80 GB SSD
PHP	8.1	8.2+
MySQL	5.7	8.0+
Web Server	Apache 2.4	Apache 2.4 + Nginx (reverse proxy)

Required PHP Extensions

```
php8.2-cli php8.2-fpm php8.2-mysql php8.2-curl php8.2-gd
php8.2-mbstring php8.2-xml php8.2-zip php8.2-bcmath php8.2-soap
php8.2-intl php8.2-readline php8.2-openssl php8.2-redis
```

Domain & DNS

- Domain registered (e.g., vueports.co.za)
- A record pointing to server IP
- www CNAME or A record configured

- admin.vueports.co.za subdomain (optional, for admin isolation)

Accounts & Keys

- PayFast merchant account (live credentials)
- Google reCAPTCHA keys (v3)
- Gmail / SMTP app password
- Google Calendar API credentials (optional)
- AWS S3 keys (for backups, optional)

2. Server Provisioning

2.1 Create Non-Root User

```
# SSH as root
ssh root@YOUR_SERVER_IP

# Create deploy user
adduser deploy
usermod -aG sudo deploy

# Copy SSH key (from local machine)
ssh-copy-id deploy@YOUR_SERVER_IP

# Disable root login & password auth
sed -i 's/PermitRootLogin yes/PermitRootLogin no/' /etc/ssh/sshd_config
sed -i 's/#PasswordAuthentication yes/PasswordAuthentication no/' /etc/ssh/sshd_config
systemctl restart sshd
```

2.2 Firewall (UFW)

```
ufw default deny incoming
ufw default allow outgoing
ufw allow ssh
ufw allow http
ufw allow https
ufw allow 8080/tcp # WebSocket server (if used)
ufw enable
```

2.3 Timezone

```
timedatectl set-timezone Africa/Johannesburg
timedatectl set-ntp yes
```

3. LAMP Stack Installation

3.1 Update System

```
sudo apt update && sudo apt upgrade -y
```

```
sudo apt install -y software-properties-common curl wget unzip git
```

3.2 Install Apache

```
sudo apt install -y apache2
sudo systemctl enable apache2
sudo a2enmod rewrite headers deflate expires ssl http2
sudo systemctl restart apache2
```

3.3 Install MySQL 8.0

```
sudo apt install -y mysql-server
sudo systemctl enable mysql

# Secure installation
sudo mysql_secure_installation

# Set root password, remove anonymous users, disallow remote root, remove test db, reload privileges
```

3.4 Install PHP 8.2 (Ondrej PPA)

```
sudo add-apt-repository -y ppa:ondrej/php
sudo apt update
sudo apt install -y php8.2-cli php8.2-fpm php8.2-mysql php8.2-curl
    php8.2-gd php8.2-mbstring php8.2-xml php8.2-zip php8.2-bcmath
    php8.2-soap php8.2-intl php8.2-readline php8.2-openssl
    php8.2-redis php8.2-imagick

# Verify
php -v # Should show PHP 8.2.x
```

3.5 Configure PHP-FPM

```
sudo nano /etc/php/8.2/fpm/pool.d/www.conf

# Change:
; user = www-data → user = deploy
; group = www-data → group = deploy
; listen.owner = www-data → listen.owner = deploy
; listen.group = www-data → listen.group = deploy

sudo nano /etc/php/8.2/fpm/php.ini

# Edit these values:
upload_max_filesize = 10M
post_max_size = 10M
max_execution_time = 300
memory_limit = 256M
max_input_vars = 3000
session.gc_maxlifetime = 1440
session.cookie_httponly = 1
session.cookie_secure = 1
session.use_strict_mode = 1
```

```
# OPcache (production)
opcache.enable = 1
opcache.memory_consumption = 128
opcache.interned_strings_buffer = 16
opcache.max_accelerated_files = 10000
opcache.revalidate_freq = 2
opcache.fast_shutdown = 1
opcache.validate_timestamps = 0 # Set to 1 for development

sudo systemctl restart php8.2-fpm
```

4. Database Setup

4.1 Create Database & User

```
sudo mysql -u root -p
```

```
CREATE DATABASE vueports_db CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
CREATE USER 'vueports_user'@'localhost' IDENTIFIED BY 'YOUR_STRONG_PASSWORD_HERE';
GRANT ALL PRIVILEGES ON vueports_db.* TO 'vueports_user'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

4.2 Import Schema

```
# Upload database.sql to server
scp database.sql deploy@YOUR_SERVER_IP:/tmp/

# Import
mysql -u vueports_user -p vueports_db < /tmp/database.sql
```

4.3 MySQL Hardening

```
sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf
```

```
[mysqld]
bind-address = 127.0.0.1
max_connections = 100
wait_timeout = 600
interactive_timeout = 600
max_allowed_packet = 16M
innodb_buffer_pool_size = 512M
innodb_log_file_size = 128M
slow_query_log = 1
slow_query_log_file = /var/log/mysql/slow.log
long_query_time = 2
```

```
sudo systemctl restart mysql
```

5. Application Deployment

5.1 Directory Structure

```
sudo mkdir -p /var/www/vueports
sudo chown deploy:deploy /var/www/vueports
```

5.2 Deploy Code

```
# Option A: Git clone (recommended)
cd /var/www/vueports
git clone https://github.com/byprojecthub/vueports-solutions.git .

# Option B: SCP upload
cd /path/to/local/project
scp -r . deploy@YOUR_SERVER_IP:/var/www/vueports/

# Set permissions
cd /var/www/vueports
find . -type f -exec chmod 644 {} \;
find . -type d -exec chmod 755 {} \;
chmod 750 config/
chmod 750 includes/
chmod 750 admin/
chmod 750 client/
chmod 750 api/
chmod 750 webhooks/
chmod 750 cron/
chmod 750 websocket/
chmod 750 logs/
chmod 750 cache/
chmod 750 uploads/
chmod 750 uploads/invoices/
chmod 750 uploads/projects/
chmod 750 uploads/avatars/

# Writable directories
chmod 775 logs/
chmod 775 cache/
chmod 775 cache/ratelimit/
chmod 775 uploads/
chmod 775 uploads/invoices/
chmod 775 uploads/projects/
chmod 775 uploads/avatars/

# Set ownership
sudo chown -R deploy:www-data /var/www/vueports
sudo chmod g+s /var/www/vueports/uploads/
sudo chmod g+s /var/www/vueports/logs/
```

```
sudo chmod g+s /var/www/vueports/cache/
```

5.3 Composer Dependencies

```
cd /var/www/vueports
composer install --no-dev --optimize-autoloader --no-interaction
```

6. Environment Configuration

6.1 Create .env File

```
cd /var/www/vueports
cp .env.example .env
chmod 600 .env
nano .env
```

```
# =====
# Vueports Solutions — Production Environment
# =====

# Application
APP_NAME="Vueports Solutions"
APP_URL=https://vueports.co.za
APP_ENV=production
APP_DEBUG=false

# Database
DB_HOST=localhost
DB_NAME=vueports_db
DB_USER=vueports_user
DB_PASS=YOUR_STRONG_PASSWORD_HERE

# SMTP / Email (Gmail App Password)
SMTP_HOST=smtp.gmail.com
SMTP_USER=njabulod.hlongwane@gmail.com
SMTP_PASS=YOUR_GMAIL_APP_PASSWORD
SMTP_PORT=587
SMTP_FROM=noreply@vueports.co.za
SMTP_FROM_NAME="Vueports Solutions"

# PayFast Payment Gateway (LIVE)
PAYFAST_MERCHANT_ID=YOUR_LIVE_MERCHANT_ID
PAYFAST_MERCHANT_KEY=YOUR_LIVE_MERCHANT_KEY
PAYFAST_PASSPHRASE=YOUR_PASSPHRASE
PAYFAST_SANDBOX=false
PAYFAST_URL=https://www.payfast.co.za/eng/process
PAYFAST_ITN_URL=https://www.payfast.co.za/eng/query/validate

# reCAPTCHA v3
```

```

RECAPTCHA_SITE_KEY=YOUR_SITE_KEY
RECAPTCHA_SECRET_KEY=YOUR_SECRET_KEY

# JWT
JWT_SECRET=GENERATE_64_CHAR_RANDOM_STRING_HERE
JWT_EXPIRY=3600

# Encryption
ENCRYPTION_KEY=GENERATE_32_CHAR_STRING_HERE

# Redis / Cache (optional)
REDIS_HOST=127.0.0.1
REDIS_PORT=6379
REDIS_PASSWORD=
CACHE_DRIVER=file

# Session
SESSION_LIFETIME=120
SESSION_SECURE=true
SESSION_HTTPONLY=true
SESSION_SAMESITE=Strict

# Rate Limiting
RATE_LIMIT_ENABLED=true
RATE_LIMIT_REQUESTS=60
RATE_LIMIT_WINDOW=60

# Backup
BACKUP_S3_BUCKET=vueports-backups
BACKUP_S3_REGION=af-south-1
AWS_ACCESS_KEY=YOUR_AWS_KEY
AWS_SECRET_KEY=YOUR_AWS_SECRET

# Monitoring
HEALTH_CHECK_KEY=random_string_for_monitoring
ERROR_REPORTING_EMAIL=njabulod.hlongwane@gmail.com

# WebSocket
WEBSOCKET_HOST=0.0.0.0
WEBSOCKET_PORT=8080
WEBSOCKET_SSL=false

# Zapier
ZAPIER_API_KEY=GENERATE_RANDOM_API_KEY_HERE

```

6.2 Generate Cryptographically Secure Keys

```

# JWT Secret (64 chars)
openssl rand -base64 48 | tr -d '=/
' | cut -c1-64

```

```
# Encryption Key (32 chars)
openssl rand -base64 24 | tr -d '=+/'
' | cut -c1-32

# Health Check Key
openssl rand -hex 16

# Zapier API Key
openssl rand -hex 32
```

7. SSL & HTTPS

7.1 Certbot (Let's Encrypt)

```
sudo apt install -y certbot python3-certbot-apache
sudo certbot --apache -d vueports.co.za -d www.vueports.co.za

# Auto-renewal test
sudo certbot renew --dry-run

# Auto-renewal is installed via systemd timer
certbot certificates
```

7.2 Apache SSL Configuration

```
sudo nano /etc/apache2/sites-available/vueports-ssl.conf
```

```
<IfModule mod_ssl.c>
<VirtualHost *:443>
    ServerName vueports.co.za
    ServerAlias www.vueports.co.za
    DocumentRoot /var/www/vueports

    # Logging
    ErrorLog ${APACHE_LOG_DIR}/vueports-error.log
    CustomLog ${APACHE_LOG_DIR}/vueports-access.log combined

    # PHP-FPM
    <FilesMatch "\.php$">
        SetHandler "proxy:unix:/run/php/php8.2-fpm.sock|fcgi://localhost"
    </FilesMatch>

    # Directory
    <Directory /var/www/vueports>
        Options -Indexes +FollowSymLinks
        AllowOverride All
        Require all granted
    </Directory>
```



```

# Security Headers
Header always set X-Frame-Options "DENY"
Header always set X-Content-Type-Options "nosniff"
Header always set X-XSS-Protection "1; mode=block"
Header always set Referrer-Policy "strict-origin-when-cross-origin"
Header always set Permissions-Policy "geolocation=(), microphone=(), camera()"
Header always set Content-Security-Policy "default-src 'self'; script-src 'self' 'unsafe-inline'
↳ https://cdnjs.cloudflare.com https://fonts.googleapis.com https://www.google.com https://
↳ www.gstatic.com; style-src 'self' 'unsafe-inline' https://fonts.googleapis.com https://
↳ cdnjs.cloudflare.com; font-src 'self' https://fonts.gstatic.com https://cdnjs.cloudflare.com;
↳ img-src 'self' https://images.unsplash.com data: blob:; connect-src 'self'; frame-src https://
↳ www.google.com; frame-ancestors 'none'; base-uri 'self'; form-action 'self';"
Header always set Strict-Transport-Security "max-age=31536000; includeSubDomains; preload"

# Compression
<IfModule mod_deflate.c>
    AddOutputFilterByType DEFLATE text/html text/css text/javascript application/javascript
    ↳ application/json
</IfModule>

# Caching
<IfModule mod_expires.c>
    ExpiresActive On
    ExpiresByType image/jpeg "access plus 1 year"
    ExpiresByType image/png "access plus 1 year"
    ExpiresByType image/svg+xml "access plus 1 year"
    ExpiresByType text/css "access plus 1 month"
    ExpiresByType text/javascript "access plus 1 month"
    ExpiresByType application/javascript "access plus 1 month"
</IfModule>

# SSL
SSLEngine on
SSLCertificateFile /etc/letsencrypt/live/vueports.co.za/fullchain.pem
SSLCertificateKeyFile /etc/letsencrypt/live/vueports.co.za/privkey.pem
SSLProtocol all -SSLv3 -TLSv1 -TLSv1.1
SSLCipherSuite
↳ ECDHE-ECDSA-AES128-GCM-SHA256:ECDHE-RSA-AES128-GCM-SHA256:ECDHE-ECDSA-AES256-GCM-SHA384:ECDHE-RSA-AES256-GCM-SHA384
SSLHonorCipherOrder on
SSLCompression off

# OCSP Stapling
SSLUseStapling on
SSLStaplingCache "shmcb:logs/stapling-cache(150000)"
</VirtualHost>
</IfModule>

```

7.3 HTTP → HTTPS Redirect

```

<VirtualHost *:80>
    ServerName vueports.co.za

```

```
ServerAlias www.vueports.co.za
Redirect permanent / https://vueports.co.za/
</VirtualHost>
```

```
sudo a2ensite vueports-ssl
sudo a2ensite vueports-redirect
sudo a2dissite 000-default
sudo systemctl restart apache2
```

8. Security Hardening

8.1 File Permissions

```
# Critical files
chmod 600 /var/www/vueports/.env
chmod 600 /var/www/vueports/config/google-credentials.json
chmod 600 /var/www/vueports/config/mailer.php
```

```
# Prevent .env and config access
sudo nano /etc/apache2/conf-available/security.conf
```

```
# Deny access to sensitive files
<FilesMatch "^(\.env|\.git|composer\.json|composer\.lock|docker-compose\.yaml|Dockerfile)$">
    Require all denied
</FilesMatch>

<DirectoryMatch "^/var/www/vueports/(config|includes|cron|websocket)/">
    Require all denied
</DirectoryMatch>
```

8.2 Fail2Ban

```
sudo apt install -y fail2ban
sudo nano /etc/fail2ban/jail.local
```

```
[DEFAULT]
bantime = 3600
findtime = 600
maxretry = 5

[apache-auth]
enabled = true

[apache-badbots]
enabled = true

[apache-noscript]
enabled = true

[apache-overflows]
```

```

enabled = true

[apache-nohome]
enabled = true

[apache-botsearch]
enabled = true

[apache-fakegooglebot]
enabled = true

[apache-modsecurity]
enabled = true

[apache-shellshock]
enabled = true

[vueports-login]
enabled = true
port = http,https
filter = vueports-login
logpath = /var/log/apache2/vueports-error.log
maxretry = 5
bantime = 3600

```

```
sudo nano /etc/fail2ban/filter.d/vueports-login.conf
```

```

[Definition]
failregex = ^.*Failed login attempt.*from <HOST>.*$
           ^.*Invalid credentials.*from <HOST>.*$
ignoreregex =

```

```

sudo systemctl restart fail2ban
sudo fail2ban-client status

```

8.3 ModSecurity (WAF)

```

sudo apt install -y libapache2-mod-security2
sudo cp /etc/modsecurity/modsecurity.conf-recommended /etc/modsecurity/modsecurity.conf
sudo sed -i 's/SecRuleEngine DetectionOnly/SecRuleEngine On/' /etc/modsecurity/modsecurity.conf
sudo systemctl restart apache2

```

8.4 ClamAV (Malware Scan)

```

sudo apt install -y clamav clamav-daemon
sudo freshclam
sudo systemctl enable clamav-daemon

# Daily scan cron
sudo nano /etc/cron.daily/clamscan

```

```
#!/bin/bash
clamscan -r /var/www/vueports/uploads --infected --remove=no --log=/var/log/clamav/daily.log

sudo chmod +x /etc/cron.daily/clamscan
```

9. Performance Optimization

9.1 Apache MPM & Tuning

```
sudo a2dismod mpm_prefork
sudo a2enmod mpm_event
sudo nano /etc/apache2/mods-available/mpm_event.conf
```

```
<IfModule mpm_event_module>
    StartServers          3
    MinSpareThreads       25
    MaxSpareThreads       75
    ThreadLimit            64
    ThreadsPerChild        25
    MaxRequestWorkers      400
    MaxConnectionsPerChild 0
</IfModule>
```

9.2 PHP-FPM Pool Tuning

```
sudo nano /etc/php/8.2/fpm/pool.d/www.conf
```

```
pm = dynamic
pm.max_children = 50
pm.start_servers = 5
pm.min_spare_servers = 5
pm.max_spare_servers = 35
pm.max_requests = 500
request_terminate_timeout = 300
```

9.3 Redis (Optional but Recommended)

```
sudo apt install -y redis-server
sudo nano /etc/redis/redis.conf
# Set: maxmemory 256mb
# Set: maxmemory-policy allkeys-lru
# Set: bind 127.0.0.1
# Set: requirepass YOUR_REDIS_PASSWORD

sudo systemctl enable redis-server
sudo systemctl restart redis-server

# Update .env
```

```
# REDIS_HOST=127.0.0.1
# REDIS_PORT=6379
# REDIS_PASSWORD=YOUR_REDIS_PASSWORD
# CACHE_DRIVER=redis
```

9.4 Asset Optimization

```
# Minify CSS (run locally before deploy)
npm install -g clean-css-cli
cleancss -o assets/css/style.min.css assets/css/*.css

# Enable in .htaccess or Apache config
# AddOutputFilterByType DEFLATE text/css text/javascript
```

10. Cron Jobs Setup

10.1 Crontab for deploy user

```
crontab -e
```

```
# Vueports Solutions — Automated Jobs
SHELL=/bin/bash
PATH=/usr/local/sbin:/usr/local/bin:/sbin:/bin:/usr/sbin:/usr/bin
MAILTO=""

# Daily database + file backup at 2:00 AM
0 2 * * * cd /var/www/vueports && php cron/daily-backup.php >> /var/log/vueports/backup.log 2>&1

# Process email queue every 5 minutes
*/5 * * * * cd /var/www/vueports && php cron/process-emails.php >> /var/log/vueports/email.log 2>&1

# Regenerate sitemap at 3:00 AM daily
0 3 * * * cd /var/www/vueports && php cron/generate-sitemap.php >> /var/log/vueports/sitemap.log 2>&1

# Subscription renewals at 4:00 AM daily
0 4 * * * cd /var/www/vueports && php cron/subscription-renewal.php >> /var/log/vueports/
↳ subscriptions.log 2>&1

# Clean old logs weekly (Sundays 5 AM)
0 5 * * 0 find /var/www/vueports/logs -name "*.log" -mtime +30 -delete

# Health check ping every 10 minutes (optional external monitoring)
*/10 * * * * curl -s -o /dev/null https://vueports.co.za/monitoring/health-check.php
```

10.2 Log Directory

```
sudo mkdir -p /var/log/vueports
sudo chown deploy:www-data /var/log/vueports
sudo chmod 775 /var/log/vueports
```

10.3 Logrotate

```
sudo nano /etc/logrotate.d/vueports
```

```
/var/log/vueports/*.log {
    daily
    missingok
    rotate 14
    compress
    delaycompress
    notifempty
    create 0644 deploy www-data
    sharedscripts
    postrotate
        /usr/bin/systemctl reload apache2 > /dev/null 2>&1 || true
    endscript
}
```

11. Monitoring & Alerts

11.1 Health Check Endpoint

Test: <https://vueports.co.za/monitoring/health-check.php>

Expected JSON response:

```
{
  "status": "healthy",
  "timestamp": "2026-05-27T14:30:00+02:00",
  "version": "1.0.0",
  "checks": {
    "database": {"status": "ok", "latency_ms": 2.15},
    "disk": {"status": "ok", "used_percent": 34.2, "free_gb": 52.8},
    "php_extensions": {"status": "ok", "missing": []},
    "memory": {"status": "ok", "usage_mb": 45.2, "limit": "256M"},
    "write_permissions": {"status": "ok", "issues": []},
    "ssl": {"status": "ok", "protocol": "TLSv1.3"},
    "env": {"status": "ok", "file": "../.env"}
  }
}
```

11.2 Uptime Monitoring (External)

Recommended services:

- **UptimeRobot** — Free tier, 5-minute checks, email/SMS alerts
- **Better Uptime** — Status page + incident management
- **Pingdom** — Detailed performance metrics

Configure to ping:

- <https://vueports.co.za/monitoring/health-check.php> (200 = healthy, 503 = unhealthy)
- <https://vueports.co.za/> (homepage)
- <https://vueports.co.za/admin/login.php> (admin portal)
- <https://vueports.co.za/client/login.php> (client portal)

11.3 Error Reporting

```
# Install Sentry PHP SDK (optional)
composer require sentry/sdk --no-dev
```

Or use simple email alerts in `includes/functions.php`:

```
set_error_handler(function($severity, $message, $file, $line) {
    if ($severity === E_ERROR || $severity === E_WARNING) {
        error_log("ERROR: $message in $file:$line");
        // Email admin on critical errors
        if (getenv('APP_ENV') === 'production') {
            mail(getenv('ERROR_REPORTING_EMAIL'), 'Vueports Error', "$message\n$file:$line");
        }
    }
});
```

12. Backup Strategy

12.1 Automated Daily Backups

Already configured via `cron/daily-backup.php`:

- MySQL dump via `mysqldump`
- Uploads directory archived
- 14-day local retention
- Optional S3 upload to `af-south-1`

12.2 Manual Backup Commands

```
# Database only
mysqldump -u vueports_user -p vueports_db > backup_$(date +%Y%m%d).sql

# Full application
tar -czf vueports_backup_$(date +%Y%m%d).tar.gz /var/www/vueports --exclude='*.log' --exclude='cache/*'

# Uploads only
tar -czf uploads_backup_$(date +%Y%m%d).tar.gz /var/www/vueports/uploads/
```

12.3 Restore Procedure

```
# Database restore
mysql -u vueports_user -p vueports_db < backup_20260527.sql
```

```
# File restore
cd /var/www
tar -xzf /path/to/vueports_backup_20260527.tar.gz
sudo chown -R deploy:www-data /var/www/vueports
```

12.4 Off-Site Backup (AWS S3)

```
# Install AWS CLI
sudo apt install -y awscli
aws configure

# Sync backups
aws s3 sync /var/www/vueports/backups/ s3://vueports-backups/production/ --storage-class STANDARD_IA

# Or add to cron/daily-backup.php (already scaffolded)
```

13. Troubleshooting

13.1 Common Issues

Symptom	Cause	Fix
500 Internal Server Error	PHP fatal error	Check <code>/var/log/apache2/vueports-error.log</code>
403 Forbidden	File permissions	<code>sudo chown -R deploy:www-data /var/www/vueports</code>
PayFast ITN not updating	IP blocked / signature fail	Check <code>logs/itn_raw.txt</code> , verify passphrase
Emails not sending	SMTP auth / app password	Verify Gmail app password, check <code>email_queue</code> table
Session expires quickly	<code>session.gc_maxlifetime</code>	Increase in <code>php.ini</code> , check <code>SESSION_LIFETIME</code> in <code>.env</code>
Slow page loads	No OPcache / no compression	Enable <code>mod_deflate</code> , verify OPcache settings
Database connection failed	Wrong credentials / MySQL down	Check <code>.env DB_*</code> vars, <code>sudo systemctl status mysql</code>
Upload fails	<code>upload_max_filesize</code>	Increase in <code>php.ini</code> and <code>.htaccess</code>
WebSocket not connecting	Firewall / port blocked	<code>sudo ufw allow 8080/tcp</code>

13.2 Diagnostic Commands

```
# Apache status
sudo systemctl status apache2
sudo apache2ctl configtest
sudo tail -f /var/log/apache2/error.log

# PHP-FPM status
sudo systemctl status php8.2-fpm
sudo tail -f /var/log/php8.2-fpm.log

# MySQL status
sudo systemctl status mysql
sudo mysql -u vueports_user -p -e "SHOW PROCESSLIST;"

# Disk space
df -h

# Memory
free -h

# Active connections
sudo netstat -tulpn | grep :80
sudo netstat -tulpn | grep :443

# Recent errors
sudo tail -n 50 /var/log/apache2/vueports-error.log
sudo tail -n 50 /var/www/vueports/logs/itn_raw.txt
```

13.3 PayFast ITN Debugging

```
# Check raw ITN logs
sudo tail -f /var/www/vueports/logs/itn_raw.txt

# Test signature generation
php -r "require '/var/www/vueports/includes/payfast.php'; var_dump(getPayFastConfig());"

# Verify IP is from PayFast
# Valid PayFast IPs: 197.97.145.144/28, 41.74.179.192/27
curl -s https://www.payfast.co.za/eng/query/validate
```

14. Post-Deployment Checklist

Immediate (Day 1)

- Homepage loads over HTTPS with valid SSL certificate
- Admin login works at /admin/login.php — **change default password immediately**
- Client registration and login work at /client/register.php
- Contact form submits and sends email

- Calculator estimates display correctly
- Booking form creates database entries
- PayFast sandbox payment completes end-to-end
- ITN webhook receives and processes test payment
- Sitemap accessible at `/sitemap.xml`
- Robots.txt accessible at `/robots.txt`
- Favicon displays in browser tab
- OG image appears in social sharing previews

Week 1

- Switch PayFast to live credentials and test R1 transaction
- Configure Google reCAPTCHA with production keys
- Set up UptimeRobot or equivalent monitoring
- Verify daily backup files are created in `/var/www/vueports/backups/`
- Test email queue processing every 5 minutes
- Review Fail2Ban logs for any intrusion attempts
- Run ClamAV scan on uploads directory
- Check Google Search Console for crawl errors
- Submit sitemap to Google and Bing

Month 1

- Review and rotate JWT secret (optional)
- Analyze `page_views` table for traffic patterns
- Check `audit_log` for any suspicious activity
- Review `consultations` conversion rates
- Update `blog_posts` with first articles
- Test disaster recovery — restore from backup to staging
- Review and optimize slow queries via MySQL slow log
- Update all packages: `sudo apt update && sudo apt upgrade`

Quick Reference Card

```
# Essential commands
sudo systemctl restart apache2 php8.2-fpm mysql
sudo tail -f /var/log/apache2/vueports-error.log
sudo tail -f /var/www/vueports/logs/itn_raw.txt
mysql -u vueports_user -p vueports_db

# File locations
App:      /var/www/vueports/
Logs:     /var/log/apache2/ + /var/log/vueports/
Backups:  /var/www/vueports/backups/
Config:   /var/www/vueports/.env
SSL:      /etc/letsencrypt/live/vueports.co.za/
PHP:      /etc/php/8.2/fpm/php.ini
Apache:   /etc/apache2/sites-available/vueports-ssl.conf
```

```
MySQL:      /etc/mysql/mysql.conf.d/mysqld.cnf
Cron:       crontab -u deploy -e
```

End of Guide

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